DESIGN AND FABRICATION OF THERMAL HEAT RESISTANCE COATING USING AN AUTONOMOUS MECHANISM IN HEAT FURNACE

A PROJECT REPORT

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of

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IN

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BONAFIDE CERTIFICATE

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ABSTRACT

In Industries where heat furnaces play a crucial role in various processes, ensuring optimal performance and longevity of furnace components is paramount. The constant exposure to high temperatures can lead to thermal degradation and structural damage, reducing the efficiency and lifespan of these essential systems. This research focuses on addressing this challenge through the development of a novel Thermal Heat Resistance Coating (THRC) integrated with an autonomous mechanism for enhanced performance and durability. The proposed THRC is designed to provide an effective barrier against extreme temperatures, minimizing heat transfer to the underlying furnace components. This coating is formulated with advanced heat-resistant materials that exhibit exceptional thermal stability and resistance to thermal cycling. Additionally, the autonomous mechanism is incorporated to actively monitor and regulate the coating's performance based on real-time temperature fluctuations within the furnace. The fabrication process involves precision application of the THRC onto the internal surfaces of the heat furnace. Special attention is given to the uniformity and thickness of the coating to guarantee consistent performance.

ACKNOWLEDMENT

At this juncture, we take the opportunity to convey our sincere thanks and gratitude to the management of the college for providing all the facilities to us.

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COMPLETION CERTIFICATE

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have completed their project on **"Design and Fabrication of Heat Resistant Coating Using an Autonomous Mechanism in Heat Furnace"** at Sri Alamelu Steels, Salem-636139 from 7th December 2023 to 7th March 2024.

During their work here they have shown satisfactory results and keen interest in the project with regular attendance.

With warm regards

Bhaskaran A

Managing partner

SRI ALAMELU STEELS SALEM 636 139

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	**iii**
	ACKNOWLEDGEMENT	**iv**
	PROJECT COMPLETION CERTIFICATE	**v**
	LIST OF TABLES	**ix**
	LIST OF FIGURES	**x**
1	**INTRODUCTION**	1
	1.1 INTRODUCTION	1
	1.2 OBJECTIVE OF THE PROJECT WORK	2
	1.3 METHODOLOGY	2
	1.4 REPORT OUTLINE	3
2	**LITERATURE REVIEW**	4
	2.1 GAP ANALYSIS	6
	2.2 PROBLEM STATEMENT	6
3	**DESCRIPTION OF COMPONENTS**	7
	3.1 COMPONENTS	7
	3.1.1 ESP32 Module	7
	3.1.2 Paint sprayer gun	8

3.1.3 Relay 9

3.1.4 SMPS 10

3.1.5 Lead Screw 10

3.1.6 DC Gear Motor 12

3.1.7 Pillow Block Bearing 12

4 DESIGN CALCULATIONS 14

4.1 FURNACE DIMENSIONS 14

4.2 FLOW RATE 14

4.3 VOLUME OF PAINT 15

4.4 VELOCITY 16

4.5 SELECTION OF MATERIAL 16

4.6 DESIGN OF SCREW 17

4.7 HELIX ANGLE FOR VERTICAL LEAD SCREW 17

4.8 FRICTION ANGLE FOR VERTICAL LEAD SCREW 18

4.9 TORQUE CALCULATION FOR VERTICAL LEAD SCREW 18

4.10 SPEED CALCULATION FOR VERTICAL LEAD SCREW 18

4.11 POWER CALCULATION 19

4.12 BUCKLING LOAD CALCULATION 19

4.13 LOAD ON THE COLUMN 20

4.14 MOMENT OF INERTIA 20

	4.15 RADIUS OF GYRATION	20
	4.16 SLENDERNESS RATIO	20
	4.17 HORIZONTAL LEAD SCREW	21
	4.18 SELECTION OF MATERIAL FOR HORIZONTAL LEAD SCREW	22
	4.19 HELIX ANGLE FOR HORIZONTAL LEAD SCREW	22
	4.20 FRICTION ANGLE FOR HORIZONTAL LEAD SCREW	23
	4.21 TORQUE CALCULATION FOR HORIZONTAL LEAD SCREW	23
	4.22 POWER CALCULATION	24
	4.23 SPEED CALCULATION	24
	4.24 ELECTRICAL CALCULATION	24
5	**FINAL ASSEMBLY AND PART DIAGRAM**	26
	5.1 MULTIVIEW PROJECTIONS	26
	5.2 MODELLING OF THE MECHANISM	27
	5.3 CIRCUIT DIAGRAM	28
	5.4 PROTOTYPE MODEL	29
6	**COST ESTIMATION**	30
7	**CONCLUSION**	31
8	**REFERENCES**	34
	ANNEXURE	36

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
5.1	BILL OF MATERIALS	27
6.1	COST ESTIMATION	30

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
1.1	BLOCK DIAGRAM	02
3.1	ESP-32 NODE MCU	07
3.2	BUILDSKILL PAINT SPRAY GUN	08
3.3	RELAY	09
3.4	SMPS	10
3.5	LEAD SCREW	11
3.6	DC GEAR MOTOR	12
3.7	PILLOW BLOCK BEARING	13
5.1	MULTIVIEW PROJECTIONS	26
5.2	3D CAD MODEL	27
5.3	CIRCUIT DIAGRAM	28
5.4	PROTOTYPE MODEL	29
5.5	CIRCUIT BOARD	29

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

In the realm of industrial processes involving high-temperature applications, the protection and optimization of heat furnace components are paramount for sustained efficiency and longevity. Traditional methods of applying coatings to furnace interiors often lack precision and adaptability, resulting in uneven coatings and suboptimal performance. This research delves into the development of an innovative solution.

Heat furnaces serve as the backbone of numerous industrial processes, including metal smelting, heat treatment, and chemical synthesis. The internal components of these furnaces are subjected to extreme temperatures, corrosive environments, and thermal cycling, necessitating robust protective measures to ensure operational reliability.

Traditional manual coating methods and fixed spraying systems often struggle to achieve uniformity in coating thickness across furnace surfaces. This lack of precision can result in hotspots, uneven thermal insulation, and reduced efficacy of the protective coating. As a consequence, furnace components are susceptible to premature wear and degradation.

The motivation behind this research lies in the quest for a more efficient and precise coating application method. The incorporation of an automated X and Y Axis Mechanism aims to address the limitations of manual techniques by enabling controlled and programmable movements for coating application. This approach seeks to enhance the uniformity, accuracy, and overall effectiveness of the coating process.

The primary objective of this study is to design, develop, and evaluate an Automated Furnace Coating System equipped with an X and Y Axis Mechanism. This mechanism allows for programmable movements in both horizontal and vertical directions, ensuring systematic coverage of furnace surfaces with a thermal-resistant coating. The research seeks to demonstrate the advantages of automation in achieving consistent coating thickness and coverage.

1.2 OBJECTIVE OF THE PROJECT WORK

To reduce the workforce of the mankind, novel robots should be introduced to perform the thermal resistance coating in heat furnace as like labours in the betterment and fast completion process of the task.

1.3 METHODOLOGY

The figure 1.1 represents the diagram of the mechanism in which the parts or functions are represented as blocks. It shows the relationship between the blocks.

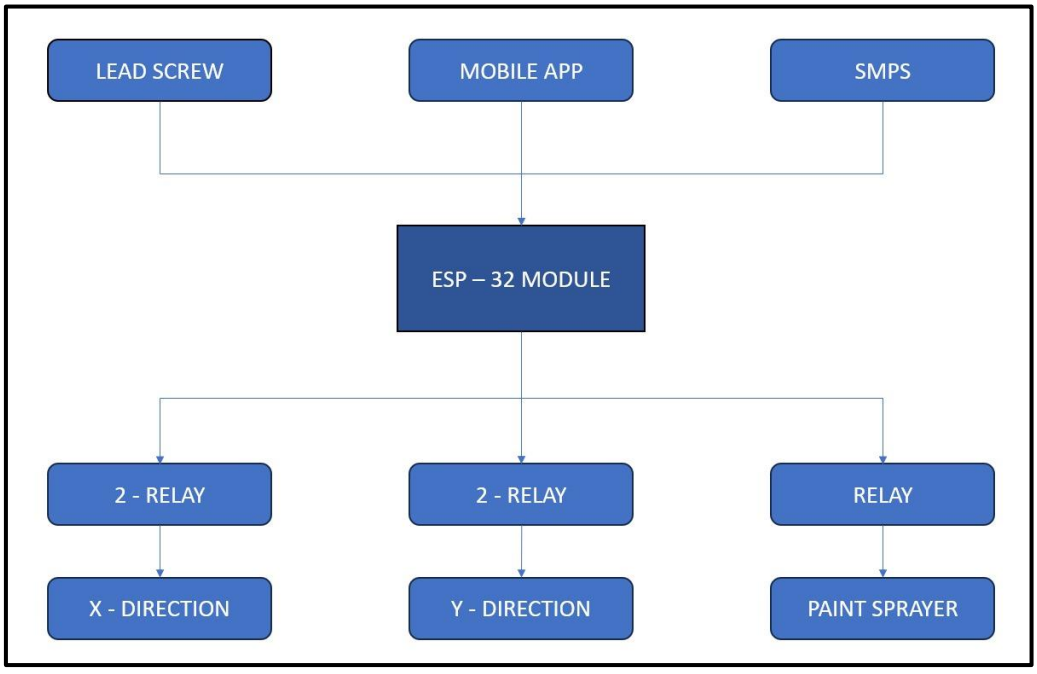


Figure 1.1 Block Diagram

1.4 REPORT OUTLINE

CHAPTER 1 discusses the basic outline of the project and where the project has been used. The outline of the project is also been discussed in this chapter.

CHAPTER 2 discusses the critical analysis and evaluation of existing research on wall painting robot and heat resistance coating. It involves identifying, analysing, and synthesizing relevant literature to gain an understanding of the current state of knowledge and identify gaps or areas for further research.

CHAPTER 3 deals with the design of the components for the autonomous mechanism. The design of bot components is a critical component of bot development. By following a structured methodology, detailed design, prototyping, iterative refinement, finalization, quality assurance, and deployment, bot components can be optimized for functionality, reliability, and cost-effectiveness.

CHAPTER 4 discusses the process of designing and developing the mechanism. By following a structured methodology that includes defining the design criteria, analysing forces and stresses, determining dimensions and materials, calculating loads and capacities, determining power requirements, and verifying the calculations, engineers can ensure that the mechanism is optimized for functionality, reliability, and safety.

CHAPTER 5 deals with the block diagram and CAD model of the mechanism. It consists of visual representation of the different components and their interconnections in a system and a digital 3D representation of the physical design of a component or system.

CHAPTER 6 deals with the cost and quantity of the resources necessary to complete a project.

CHAPTER 7 discusses the main points, findings, and implications of the research.

CHAPTER 2

LITERATURE REVIEW

It is always good to get expert opinions before starting a new project. We researched on our topic and got reference from various experts in this chapter. Below given is the list of books and publications related to our project.

Proano-Aviles *et al* [8] discussed that these reactors, combined with GC-MS or FID have been widely employed in pyrolysis research. The capability of these devices to meet the expected criteria of fast heating rates and short vapor residence times typical in fast-pyrolysis processes has not been adequately verified.

A. A. Bhat *et al* [1] discussed that the primary objective is to develop induction heating furnaces suitable for vacuum distillation processes aimed at recovering and consolidating heavy metals post electrorefining operations. A mock-up induction furnace is planned to replicate conditions anticipated in an actual induction-heated vacuum distillation furnace.

Enes Kilinç *et al* [3] discussed that the context highlights the pressing need for efficient energy resource utilization due to factors such as rapid energy consumption, escalating energy demands, industry competitiveness, and growing environmental concerns.

Keng Huat Koh *et al* [4] discussed that the primary research focus revolves around the adhesion and locomotion mechanism of wall climbing robots. The aim is to develop an efficient, robust, standalone, and scalable mechanism for these robots to perform various critical tasks.

Zhuang-nian Li et al [5] discussed that the paper introduces an efficient model for predicting and controlling temperatures in large blast furnaces. It utilizes thermal equilibrium principles and considers the carbon-oxygen balance within the high-temperature zone of the blast furnace.

Miroslav Rimar et al [9] discussed that heating flame furnaces are extensively employed in metallurgy and mechanical engineering for heating and heat treatment of metal products. Modern heating furnaces strive to maintain uniformity within the working chamber by avoiding high-temperature or stagnation zones.

Rajesh Kannan Megalingam et al [7] discussed that this mechanism assists the paint sprayer in reaching required heights on the walls, ensuring coverage across the surface efficiently. Ultrasonic sensors are utilized by the robot to detect distances from walls.

Koushik Chakravarty et al [2] discussed that reheating furnaces play a pivotal role in the iron and steel industry by reheating semi-finished products to rolling temperatures (e.g., slabs or billets at 1250 ◦C). However, they are significant energy consumers in rolling mills. Optimizing fuel consumption in reheating processes is critical to prevent overheating of steel materials.

Gandhi Mallela et al [6] rolling generators have a significant role in the metal making process, where they transform metal blooms from the melting shop into finished steel products. The blooms obtained from the metal melting shop are rapidly processed in a reheating furnace to be heated and prepared for rolling.

Utkarsha S. Bawane et al [10] discussed that a new robot designed for exterior wall painting has been developed. It is mounted on equipment that allows movement along the exterior walls of a building, with capabilities to move up, down, left, and right. The robot is computer-controlled and can be activated simply by the operator pressing a switch on the control panel located on the ground. The goal of the Automatic Wall Painting Robot is to minimize human interaction in the painting process and reduce painting equipment costs.

2.1 GAP ANALYSIS

A very few implementations have been taken over in this case of coating thermal heat resistance in heat furnace. As of now, manpower is used to achieve this specific task. By implementing our project, we can find a solution by turning this into fully automated process and this will result in a one-time investment and a long-term benefit.

2.2 PROBLEM STATEMENT

To reduce the workforce of the mankind, novel robots should be introduced to perform the thermal resistance coating in heat furnace as like labours in the betterment and fast completion process of the task.

CHAPTER 3
DESCRIPTION OF COMPONENTS

3.1 COMPONENTS

The Components used in the project are listed below with figure and brief description.

3.1.1 ESP-32 Module

ESP WROOM 32 is a powerful, generic WiFi-BT-BLE MCU module that targets a wide variety of applications, ranging from low-power sensor networks to the most demanding tasks, such as voice encoding, music streaming, and MP3 decoding. As shown in Figure 3.1 the core of this module is the ESP32S chip, which is designed to be scalable and adaptive. There are 2 CPU cores that can be individually controlled or powered, and the clock frequency is adjustable from 80 MHz to 240 MHz

Using Bluetooth, users can connect to their phone or broadcast low energy beacons for its detection. The use of Wi-Fi enables a large physical range, as well as a direct connection to the internet via a Wi-Fi router. In addition, this module can support data rates of up to 150 Mbps and 22 dBm output power at the PA in order to allow for the widest physical range.

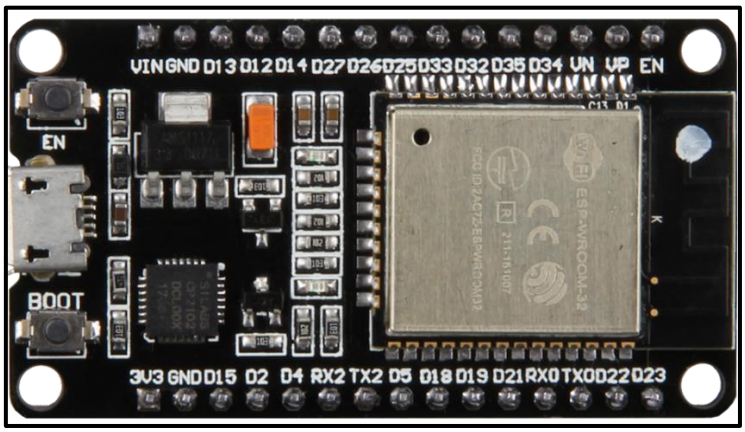


Figure 3.1 ESP-32 Node MCU

3.1.2 Paint Spray Gun

A paint spray gun is a versatile tool used for spreading and applying paint, varnish, or other coating materials onto surfaces. It operates by utilizing compressed air to atomize paint particles, directing them toward the intended surface. The figure 3.2 shows the key components of a spray gun include the nozzle, which shapes the spray pattern, the paint basin for holding the coating material, and an air compressor to provide the necessary air pressure. When the trigger is pressed, the paint mixes with the compressed air stream and is released as a fine spray onto the surface.

Spray guns utilize compressed air to atomize the paint particles and direct them toward the intended surface. Nozzle shapes the spray pattern and determines the consistency of the paint. Paint Basin holds the paint or coating material. Air Compressor provides the necessary air pressure. When the trigger is pressed, the paint mixes with the compressed air stream and is released as a fine spray onto the surface.



Figure 3.2 Buildskill Paint Spray Gun

3.1.3 Relay

Relay is an electrically operated switch. Current flowing through the coil of the relay creates a magnetic field which attracts a lever and changes the switch contacts. The coil current can be on or off so relays have two switch positions and they are double throw (changeover) switches. Relays allow one circuit to switch a second circuit which can be completely separate from the first. For example, a low voltage battery circuit can use a relay to switch a 230V AC mains circuit. There is no electrical connection inside the relay between the two circuits; the link is magnetic and mechanical. Figure 3.3 represents the standard relay.

The coil of a relay passes a relatively large current, typically 30mA for a 12V relay, but it can be as much as 100mA for relays designed to operate from lower voltages. Most ICs (chips) cannot provide this current and a transistor is usually used to amplify the small IC current to the larger value required for the relay coil.

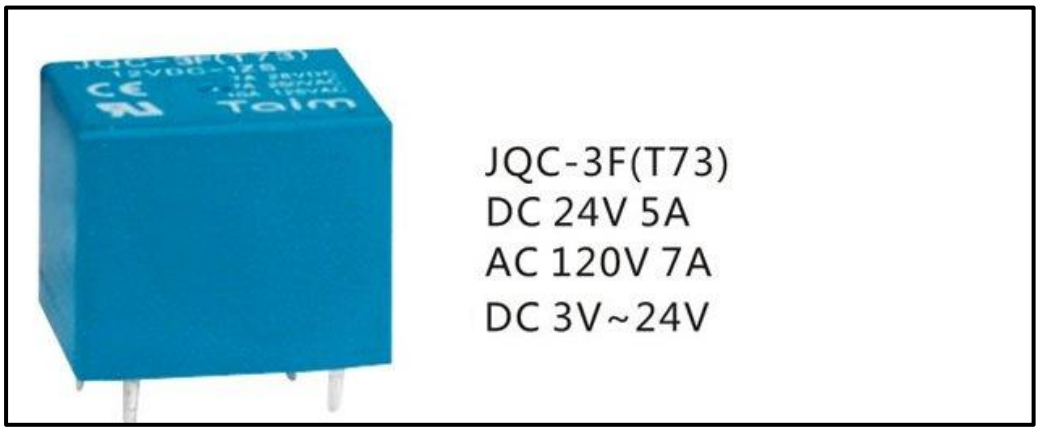


Figure 3.3 Relay

3.1.4 SMPS – Switched Mode Power Supply

A switched-mode power supply (SMPS) is an electronic circuit that converts power using switching devices that are turned on and off at high frequencies, and storage components such as inductors or capacitors to supply power when the switching device is in its non-conduction state. Switching power

supplies have high efficiency and are widely used in a variety of electronic equipment, including computers and other sensitive equipment requiring stable and efficient power supply. The figure 3.4 is a switched-mode power supply of 110V/220V. It is also known as a switch-mode power supply or switching-mode power supply.

The input DC supply from a rectifier or battery is fed to the inverter where it is turned on and off at high frequencies of between 20 KHz and 200 KHz by the switching MOSFET or power transistors. The high-frequency voltage pulses from the inverter are fed to the transformer primary winding, and the secondary AC output is rectified and smoothed to produce the required DC voltages. A feedback circuit monitors the output voltage and instructs the control circuit to adjust the duty cycle to maintain the output at the desired level.



Figure 3.4 SMPS

3.1.5 Lead Screw

Lead screw is a power transmission linkage used in modern machines. It can generate very high forces with a small moment, thus providing a large mechanical advantage. In figure 3.5 lead screw can be thought of as a wedge

wrapped around a cylindrical rod. In applications such as screw jacks and mechanical presses, it is used to create large forces. They also find use in applications needing extremely precise motion transfer and control, such as linear actuators and linear stages. As the sliding contact area between the screw and the nut is high, a lead screw has more friction losses compared to other alternatives such as gear trains and chain drives.

The screw shaft is a cylindrical rod with threads or grooves along its length. At times, it may be referred to as an ACME rod, buttress rod or square-threaded rod as per the thread geometry. Although it is not a separate component in itself, the thread's structure is responsible for converting rotational motion into linear motion. The external threads are in direct contact and mesh with the nut's internal threads. The nut is a fairly simple component but does not offer much use by itself. Typically, it will have some means, such as tapped or through holes, to connect it to supporting components like a clamp.

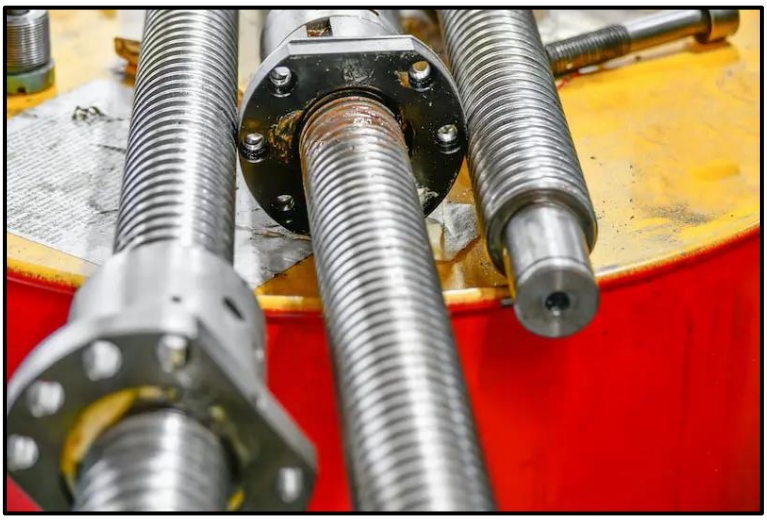


Figure 3.5 Lead Screw

3.1.6 DC Gear Motor

A DC gear motor, short for direct current gear motor, is a type of electric motor integrated with a gear train. Figure 3.6 represents the DC gear motor. This combination enables the motor to deliver increased torque and lower speed output compared to a standard DC motor. The gear train consists of gears that mesh together to reduce the rotational speed of the motor's output shaft while increasing its torque.



Figure 3.6 DC Gear Motor

3.1.7 Pillow Block Bearing

A pillow block bearing, also known as a Plummer block, is a crucial component in machinery and industrial applications. Figure 3.7 represents the pillow block bearing which provides support and stability for rotating shafts. The assembly consists of an anti-friction bearing housed within a sturdy block (the pillow block). This block is then mounted to a foundation or structure. The shaft passes through the centre of the bearing, allowing it to rotate freely. Pillow block bearings come in various types, including ball, cylindrical roller, spherical roller, and tapered roller bearings. These bearings find applications in conveyors, agricultural equipment, and other mechanical systems. The choice of material for the housing can vary, with options like cast iron, composite, pressed steel, or

stainless steel. Overall, pillow block bearings play a vital role in ensuring smooth and efficient machinery operation.

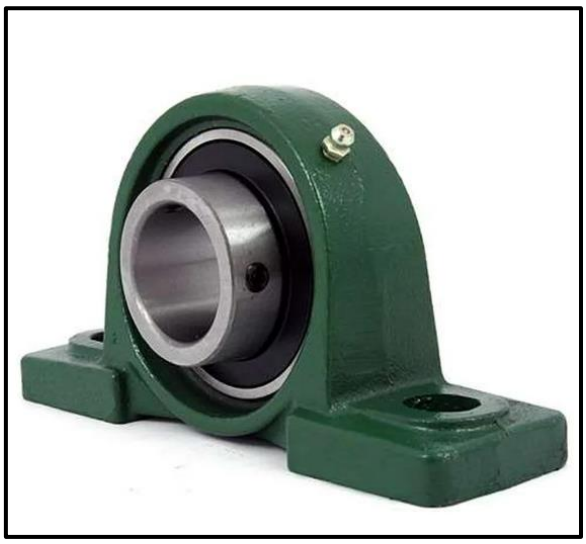


Figure 3.7 Pillow Block Bearing

CHAPTER 4
DESIGN CALCULATIONS

4.1 FURNACE DIMENSIONS

The furnace dimensions of the industry are stated below.

Length = 18 feet = 5.48 m = 5.5m

Breadth = 6.5 feet = 1.98 m = 2m

Surface Area = l*b

Surface Area = 5.5*2

Surface Area = 11 m^2

4.2 FLOW RATE

The below calculations are to select the Sprayer and nozzle.

Surface area to be painted = 11m^2

Application rate = Volume of paint/Time

Volume of paint = Surface area * Coating thickness

Coating Thickness = Thermal Conductivity = 0.000063 W/mk [Watts per meter Kelvin]

Coating Thickness = Time*Thermal Conductivity*Surface area/Temperature Difference

Temperature Difference = Max.Temperature - Ambient Temperature

Surface area to be painted = 11m^2

Application rate = Volume of paint/Time

Volume of paint = Surface area * Coating thickness

Coating Thickness = Thermal Conductivity = 0.000063 W/mk [Watts per meter Kelvin]

Coating Thickness = Time*Thermal Conductivity*Surface area/Temperature Difference

Temperature Difference = Max.Temperature - Ambient Temperature

Maximum Temperature = 1200˚ C

Ambient Temperature = 40˚ to 50˚ C = 40˚ C

Furnace Working time = 807 hrs => 33 1/2 days

4.3 VOLUME OF PAINT

The selection of sprayer requires the volume of paint used. With the help of flow rate and volume nozzle and sprayer are chosen.

Coating Thickness= 807*0.000063*11/(1200-40)

= 5559.25/1160

= 0.00048 mm

= 0.5 mm

Volume of paint = Surface area * Coating Thickness

= 11*0.5 mm

= 11 m^2 * 0.0005 m

= 0.0055 m^3

Flow rate:

= Surface Area * Application rate

= 11 m^2* 0.000091 m^3/min

= 11*0.091 l/min

= 1.001 l/min

1 l/mm nozzle Sprayer is to be Selected

Sprayer Weight = 850 gms

4.4 VELOCITY

The velocity of the sprayer is calculated below.

Area Covered by Sprayer:

Area Covered by Sprayer = 0.798 m^2

Total Area = 11 m^2

No of time up and down motion of the sprayer = 11/0.793

= 13 times

Assume time taken = 45 mins

Time taken for one up and down = 45/18

= 3.46 mins

= 207.6 secs

Velocity = Distance/Time

= 2/207.6

= 0.0096 m/s

= 0.010 m/s

4.5 SELECTION OF MATERIAL

The material for the lead screw is chosen.

Cast Iron (FG 200)

Ultimate tensile strength = 200 N/mm²

Young's Modulus = 180*10³

W =m*g

=0.85*9.8

=8.33 N

Permissible compressive stress = Ultimate Tensile Strength/FOS

= 200/4

= 50 N/mm²

4.6 DESIGN OF SCREW

Design of the vertical screw is calculated.

$50= W/((\pi/4) *dc^2))$

$50 = 8.33/((\pi/4) *dc^2))$

$(\pi/4) *dc^2)) = 8.33/50$

$d_c^2 = 1.69$

$d_c = 1.5$ mm

$d = dc/0.8$

$d = 1.5/8$

$d = 2$mm

$d_c = d-p$

$d_c = 2-1.5$

$d_c = 0.5$ mm

d_m =d-(0.5*p)

d_m = 2-(0.5*1.5)

d_m = 1.125

4.7 HELIX ANGLE FOR VERTICAL LEAD SCREW

tan α = p/π*dm

tan α = 1.5/π*1.125

α = tan-1(1.5/3.92)

α = 20.9

4.8 FRICTION ANGLE FOR VERTICAL LEAD SCREW

tan(ø)=1/2tan(β)+μ

= (1/2) *tan (30) +0.2

= tan-1(0.486)

=2 6.04

α < ø

The Screw is Self-locking.

4.9 TORQUE CALCULATION FOR VERTICAL LEAD SCREW

T = F*P/(2*π*η*μ)

= 8.33*1.5/(2*π*0.8*0.2)

T = 12.4 Nmm

4.10 SPEED CALCULATION FOR VERTICAL LEAD SCREW

V = π*D*N/60

= 0.010*60/2*π

V = 95.4 rpm

= 100 rpm

4.11 POWER CALCULATION

Power calculation of the motor which is connected in vertical lead screw.

$P = 2*\pi*N*T/60$

$= 2*\pi*100*12.4/60$

P = 129.8 W

4.12 BUCKLING LOAD CALCULATION

Buckling load for vertical lead screw is calculated.

Mass = Density * Volume

$V = \pi*(D/2)^2 * 1000$

$= \pi*(1.25/2)^2 * 1000$

$= 1.25*10^{-5}\ m^3$

$m = 6800\ kg/m^3 * 1.25*10^{-5}$

$= 6800*1.25*10^{-5}$

= 0.085 kg

Screw Mass = 0.085 kg

Sprayer Mass = 0.850 kg

Total Load = 0.850+0.085

= 0.985

Total Load = 1 kg

4.13 LOAD ON THE COLUMN

Load on the column for vertical lead screw is calculated.

The load acting on the column = 1*9.8

= 9.8 N

Standard dimension of hollow tube from the chart Section: G100196

A= (b-t) * (b-t)

= 28*28

= 784 mm^4

4.14 MOMENT OF INERTIA

Moment of inertia for vertical lead screw is calculated.

$I = [(a^4-a^4)/12]$

$= (30^4-26^4)/12$

= 29418.66 mm^2

4.15 RADIUS OF GYRATION

Radius of gyration for vertical lead screw is calculated.

$k= (I/A)^{1/2}$

$= (29418.6/784)^{1/2}$

k=6.125 mm

4.16 SLENDERNESS RATIO

Slenderness ratio for vertical lead screw is calculated.

$(l/k)_{actual}$ = 1000/6.125

= 163.247

$yt/2 = n*\pi^2 *E/(l/k)^2$

Young's Modulus = 7000 N/mm^2

Yield Stress = 503 MPA

Both ends are fixed so n=1 is taken

$l/k = (1*\pi^2*7000*2/503)^{1/2}$

$l/k = (2747.007)^{1/2}$

$(l/k)_{critical} = 52.41$

$(l/k)_{critical} < (l/k)_{actual}$ [The column is long.]

By using Euler's equation:

$P_{critical} = n*\pi^2*7000*2/(l/k)^2$

$= 1*\pi^2*7000*784/ (163.217)^2$

$= 2034.6$ N

$P_{critical} > P_{actual}$ [So, the design is safe.]

4.17 HORIZONTAL LEAD SCREW

The total weight of the frame material is Mild Steel

Weight = 0.713 kg/m

$= 0.713*10^{-3}*2600$

= 1.853 kg

Total weight = 1.853 * 4 =7.5kg

The total weight acting on horizontal Screw

= Sprayer weight + Screw Weight + Frame Weight

= (7.8538+0.850+0.85) = 9.2 kg => 10kg

The load acting on each Screw = 3 kg/2

= 1.5 kg

Weight = mass * gravity

= 1.5 *9.8

= 14.7 N

4.18 SELECTION OF MATERIAL HORIZONTAL LEAD SCREW

Cast iron (FG 200)

w = 200 N/mm^2

c = Ultimate Tensile Strength /FOS

= 200/4

= 50 N/mm^2

$c = W/((\pi/4) * dc^2))$

$50 = 14.7/((\pi/4) * dc^2))$

$(\pi/4) *dc^2)) = 14.7/50$

$d_c^2 = 1.69$

$d_c = 1.5$ mm

$d = d_c /0.8$

d = 1.5/8

d = 2mm

d_c = d-p

d_c = 2-1.5

d_c = 0.5 mm

d_m =d-(0.5*p)

d_m = 2-(0.5*1.5)

d_m = 1.125 mm

4.19 HELIX ANGLE FOR HORIZONTAL LEAD SCREW

tan α = p/π*dm

tan α = 1.5/π*1.125

α = tan-1(1.5/3.92)

α = 20.9

4.20 FRICTION ANGLE FOR HORIZONTAL LEAD SCREW

tan(ø) = 1/2tan(β)+μ

= (1/2) *tan (30) + 0.2

= tan-1(0.486)

= 26.04

α < ø

The Screw is Self-locking.

4.21 TORQUE CALCULATION FOR HORIZONTAL LEAD SCREW

T = F*P/(2*π*η*μ)

= 14.7*1.5/(2*π*0.8*0.2)

T = 21.9 Nmm => 22 Nmm

4.22 POWER CALCULATION

Power calculation of the motor which is connected in horizontal lead screw.

$P = 2*\pi*N*T/60$

$= 2*\pi*100*22/60$

$P = 230.3$ W

4.23 SPEED CALCULATION

Speed of the horizontal lead screw is calculated.

$V = \pi*D*N/60$

$= 0.010*60/2*\pi$

$V = 95.4$ rpm

$= 100$ rpm

4.24 ELECTRICAL CALCULATION

1.Components used:

i) Motors for Motion

ii) Relays for Control Switch

2. Motor Calculation:

Area Covered = 0.793 m^2

Total Area = 11 m^2

Motor rpm = 100 rpm [Calculated before]

3 Motors = 3*100

= 300 rpm

Frequency = 300/60

= 5 Hz

3. Relay Calculation:

Relay Standard Value = 5 V

4 Relays = 5*4 = 20 V

AC load current = 10 A

= 10*4

= 40 A

Frequency = 12 Hz

Power = V*I

= 20*40

= 800 W

Microcontroller Chosen:

Total Frequency Generated = 5 Hz + 12 Hz

= 17 Hz

Bit rate = 16 bit

According to the results attained from the calculations and the need for a 16-bit rate microcontroller, we chose ESP32 Microcontroller.

CHAPTER 5

FINAL ASSEMBLY AND PART DIAGRAM

5.1 MULTIVIEW PROJECTIONS

The following figure 5.1 represents the different views of the model.

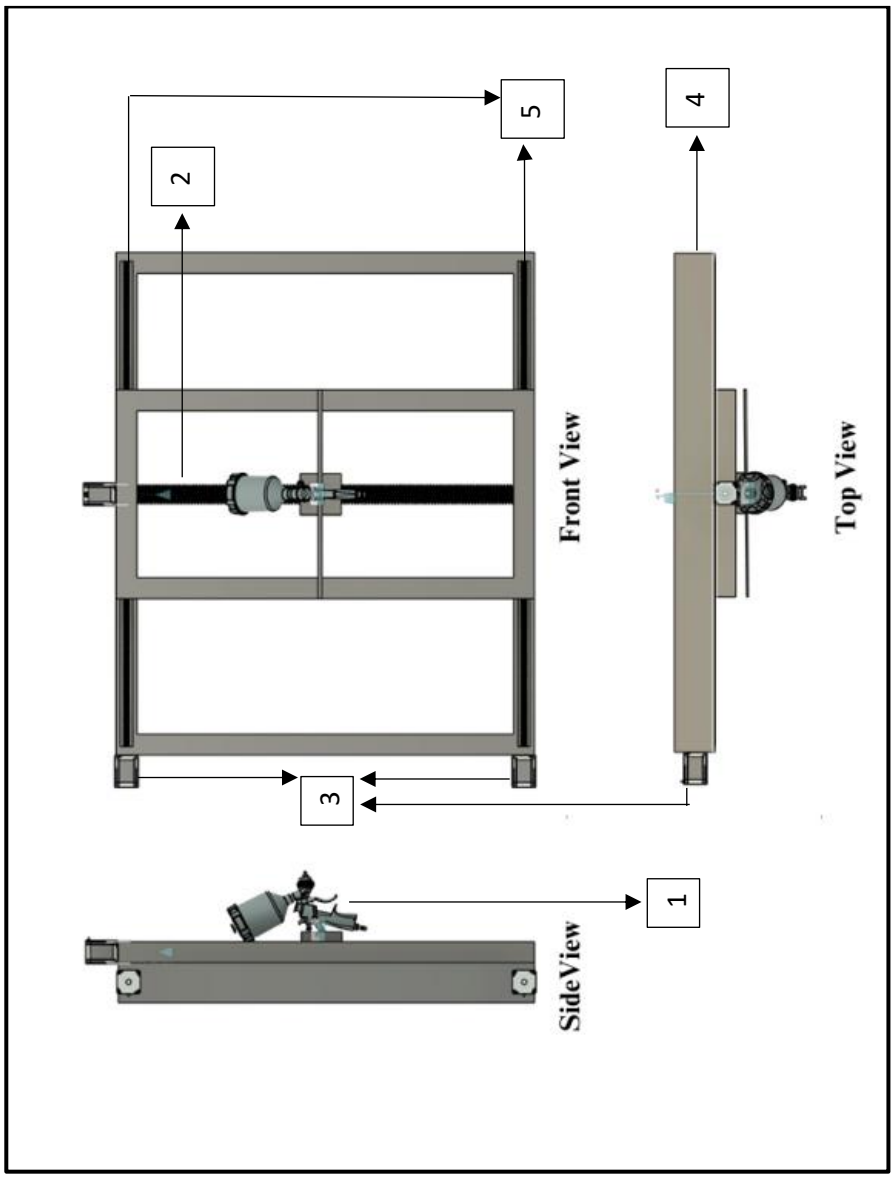


Figure 5.1 Multiview Projections

Table 5.1 Bill of Materials

The Table 5.1 represents the bill of material for the components used in project.

NUMBER	COMPONENT	MATERIAL	PRICE
1	Spray Gun	Standard	2500
2	Vertical Lead Screw	Mild Steel	800
3	DC Gear Motor	Mild Steel	2000
4	Frame	Mild Steel	2000
5	Horizontal Lead Screw	Mild Steel	800

5.2 MODELLING OF THE MECHANISM

The following figure 5.2 represents the proposed CAD model of the mechanism.

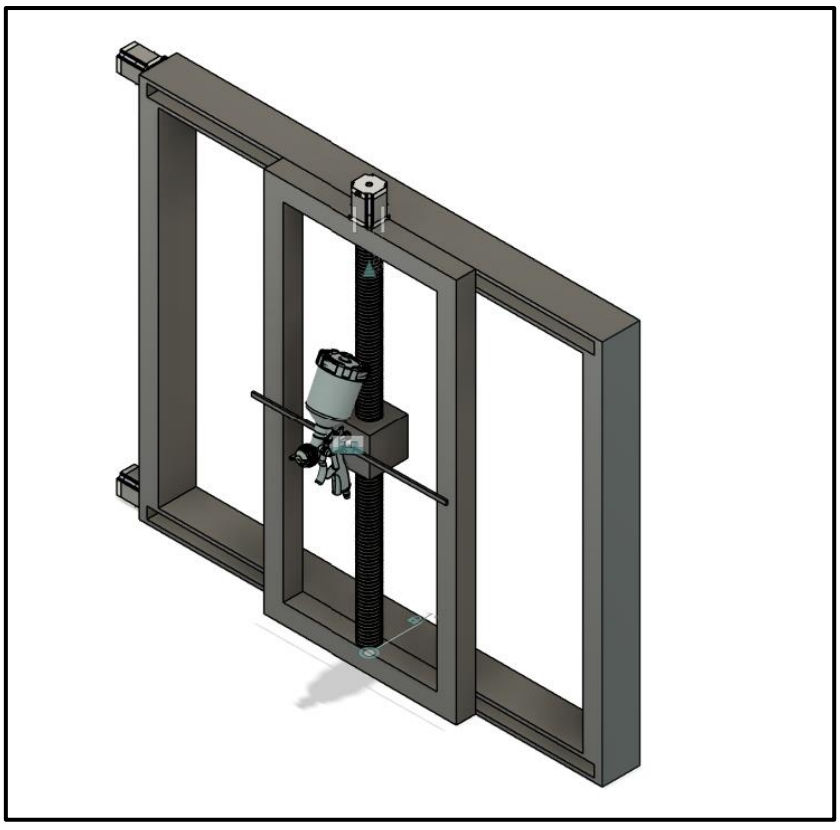


Figure 5.2 3D CAD Model

5.3 CIRCUIT DIAGRAM

The following figure 5.3 represents the circuit of the system. Various components used in the system are represented in this circuit diagram.

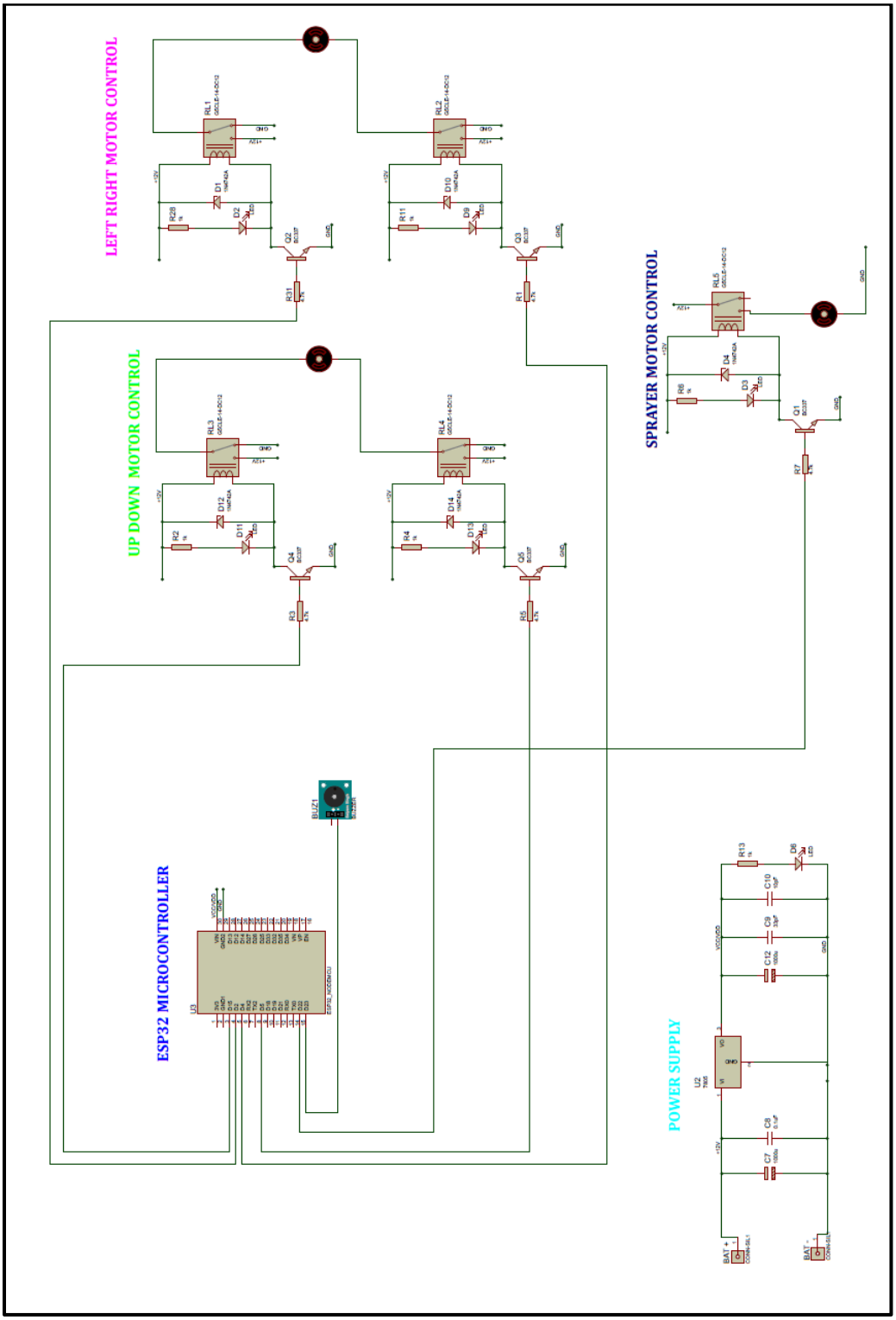


Figure 5.3 Circuit Diagram

5.4 PROTOTYPE MODEL

The figure 5.4 and figure 5.5 is the outcome of the project and the function is effectively observed.



Figure 5.4 Prototype model

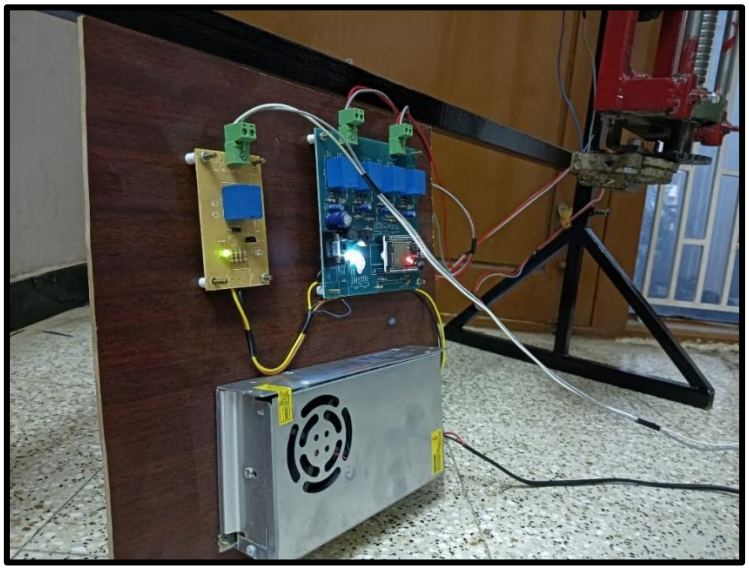


Figure 5.5 Circuit Board

CHAPTER 6

COST ESTIMATION

Table 6.1 Cost Estimation

The following Table 6.1 represents the estimated cost of each component used in the project as well as the total cost estimated to complete the project.

S.NO	COMPONENTS	QUANTITY	COST
1	DC Gear Motor	02	2000
2	Linear Guide	04	1600
3	Pillow Block Bearing	02	1200
4	Screw Rod	02	5000
5	Spray Gun	01	2500
6	ESP32	01	1000
7	Relay	05	500
8	SMPS	01	1200
9	Mild Steel Rectangular Hollow Pipe	15m	2250
10	Buzzer	01	100
11	Miscellaneous		500
12	Fabrication expenses		4500
13	Transportation expenses		1000
TOTAL			**Rs. 23,350**

So, the total expense spent on the completion of project is around Rs.23,500.

CHAPTER 7

CONCLUSION

The integration of a thermal heat resistance coating system in a heat furnace, coupled with an autonomous mechanism featuring a lead screw and ESP32 microcontroller, represents a cutting-edge and efficient solution for enhancing heat management and operational control. The application of a thermal heat resistance coating significantly improves the furnace's ability to withstand high temperatures and thermal stresses, thereby enhancing overall thermal efficiency. This translates into improved energy utilization and reduced heat loss during operation.

The incorporation of an autonomous mechanism, driven by a lead screw and controlled by an ESP32 microcontroller, introduces a high level of automation to the heat resistance coating process. This allows for precise and programmable control over the coating application, ensuring consistency and accuracy in challenging furnace environments. The autonomous system, coupled with the thermal heat resistance coating, promotes extended equipment life by minimizing wear and tear associated with extreme temperatures. Additionally, the autonomous mechanism allows for efficient maintenance scheduling and self-diagnosis, further enhancing the overall durability and reliability of the furnace.

The combination of a thermal heat resistance coating system, an autonomous mechanism with a lead screw, and the ESP32 microcontroller introduces a comprehensive and advanced solution for optimizing heat furnace performance. This integrated approach addresses key challenges associated with extreme temperatures, energy efficiency, and overall system control, making it a valuable advancement in industrial heating applications.

SCOPE FOR FUTURE WORKS

Intelligent Machine Learning Algorithms: Implement machine learning algorithms on the ESP32 microcontroller to enhance the autonomous control system. Adaptive algorithms could optimize the coating process based on historical data, enabling the system to continually improve efficiency and performance.

Remote Monitoring and Control: Explore the integration of IoT (Internet of Things) technology for remote monitoring and control of the heat furnace system. This could allow operators to monitor and adjust the coating process from a distance, improving flexibility and responsiveness in various industrial settings.

Energy Harvesting Solutions: Investigate energy harvesting technologies to power the autonomous system without relying solely on external power sources. This could involve exploring options such as solar panels or other innovative energy recovery mechanisms to make the system more sustainable and self-sufficient.

Real-Time Diagnostics and Predictive Maintenance: Implement advanced diagnostics and predictive maintenance capabilities within the ESP32 microcontroller. This could involve the development of sensors and algorithms to detect potential issues in real-time, allowing for proactive maintenance and minimizing downtime.

LIMITATIONS

Dependency on External Power: The autonomous mechanism, relying on the ESP32 microcontroller, requires a stable power source. Interruptions or failures in the power supply could disrupt the autonomous control system, affecting the coating process and furnace operation.

Initial Setup and Calibration: Calibrating the autonomous system for optimal performance and adapting it to specific furnace conditions may require expertise and time. Improper calibration could lead to suboptimal coating results or inefficient furnace operation.

Limited Adaptability: The system's adaptability to diverse furnace configurations and varying industrial environments may be limited. Customization may be required for specific applications, and the system's flexibility in accommodating different furnace designs should be considered.

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ANNEXURE

```
/*
blynk id and password
paintsprayer@gmailwe.com
1234567890
*/
#define BLYNK_TEMPLATE_ID           "TMPL3oaIRCh7D"
#define BLYNK_TEMPLATE_NAME         "Quickstart Template"
#define BLYNK_AUTH_TOKEN            "3J2-8-
DejcTW7B7uTZlz0kXW8j8BaI7G"
#define BLYNK_PRINT Serial
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
char auth[] = BLYNK_AUTH_TOKEN;
char ssid[] = "admin";
char pass[] = "1234567890";
int x,y,x1,x2,x3,x4,x5, ksts,sts1,dir,cl;
int gas,moi,sts;
#define r1  15
#define r2  2
#define r3  4
#define r4  5
```

```
#define r5  22
#define BUZZER  23
BLYNK_CONNECTED()
{
  Blynk.syncVirtual(V0);
  Blynk.syncVirtual(V1);
  Blynk.syncVirtual(V2);
  Blynk.syncVirtual(V3);
  Blynk.syncVirtual(V4);
  }
BLYNK_WRITE(V0)
{
x1 = param.asInt();
if(x1==1){sts=1;Serial.println("FWD");} // RIGHT
if(x1==0){sts=0;} // RIGHT
}
BLYNK_WRITE(V1)
{
x2 = param.asInt();
if(x2==1){sts=2;Serial.println("rev");} // RIGHT
if(x2==0){sts=0;} // RIGHT
}
```

```
BLYNK_WRITE(V2)
{
x3 = param.asInt();
if(x3==1){sts=3;Serial.println("right");} // RIGHT
if(x3==0){sts=0;} // RIGHT
}
BLYNK_WRITE(V3)
{
x4 = param.asInt();
if(x4==1){sts=4;Serial.println("left");} // RIGHT
if(x4==0){sts=0;} // RIGHT
}
BLYNK_WRITE(V4)
{
x5 = param.asInt();
if(x5==1){digitalWrite(r5,HIGH);Serial.println("SPRAYER ON");}
if(x5==0){digitalWrite(r5,LOW);Serial.println("SPRAYER OFF");}
}
void setup()
{
Serial.begin(9600);
pinMode(r1, OUTPUT);
```

```
pinMode(r2, OUTPUT);
pinMode(r3, OUTPUT);
pinMode(r4, OUTPUT);
pinMode(r5, OUTPUT);
pinMode(BUZZER, OUTPUT);

digitalWrite(r1,LOW);
digitalWrite(r2,LOW);
digitalWrite(r3,LOW);
digitalWrite(r4,LOW);
digitalWrite(r5,LOW);
   Blynk.begin(auth, ssid, pass);
   digitalWrite(BUZZER,HIGH);
   delay(1500);
   digitalWrite(BUZZER,LOW);
}
void loop()
{
  Blynk.run();
  if(sts==1){digitalWrite(r1,LOW);digitalWrite(r2,HIGH);}
  if(sts==2){digitalWrite(r1,HIGH);digitalWrite(r2,LOW);}
  if(sts==3){digitalWrite(r3,HIGH);digitalWrite(r4,LOW);}
```

```
  if(sts==4){digitalWrite(r3,LOW);digitalWrite(r4,HIGH);}
  if(sts==0){digitalWrite(r1,LOW);
digitalWrite(r2,LOW);
digitalWrite(r3,LOW);
digitalWrite(r4,LOW);
}
 }
```